



Lecture 5: Multiple discrete random variables

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1. Review
2. Conditioning
3. Multiple discrete random variables
4. Independence
5. Recitation
6. Summary



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- Random variable X : **function** from sample space to real numbers
- PMF (for discrete random variables):

$$p_X(x) = \mathbf{P}(X = x) = \mathbf{P}\{\omega : X(\omega) = x\}$$

- Expectation:

$$\mathbf{E}[X] = \sum_x x p_X(x)$$

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x)$$

$$\mathbf{E}[\alpha X + \beta] = \alpha \mathbf{E}[X] + \beta$$

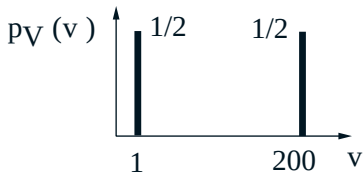
- Variance:

$$\begin{aligned}\text{var}(X) &= \mathbf{E} \left[(X - \mathbf{E}[X])^2 \right] \\ &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2\end{aligned}$$

- Standard deviation: $\sigma_X = \sqrt{\text{var}(X)}$

Average speed vs. average time

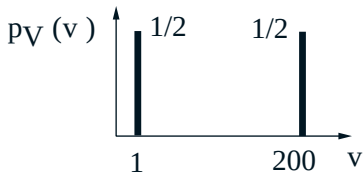
- Traverse a 200 mile distance at constant but random speed V



- $d = 200$, $T = t(V) = 200/V$
- $\mathbf{E}[V] = 100.5$
- $\mathbf{E}[T] = \mathbf{E}[t(V)] = \sum_v t(v)p_V(v) = 100.5$

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- $\mathbf{E}[T] = \mathbf{E}[t(V)] = \sum_v t(v)p_V(v) = 100.5$
- $\mathbf{E}[TV] = 200 \neq \mathbf{E}[T] \cdot \mathbf{E}[V]$

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Definition of Conditional PMF

The conditional PMF of a RV X given an event A with $\mathbf{P}(A) > 0$ is defined by

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}$$

- Since $\mathbf{P}(A) = \sum_x \mathbf{P}(\{X = x\} \cap A)$,

$$\sum_x p_{X|A}(x) = 1$$

- Conditional PMFs are similar to ordinary PMFs, but pertain to a universe where the conditioning event is known to have occurred.

Definition of Conditional Expectations

The conditional expectation of a RV X given an event A with $\mathbf{P}(A) > 0$ is defined by

$$\mathbf{E}[X \mid A] = \sum_x x p_{X|A}(x).$$

The conditional expectation of X given a value y of Y is defined by

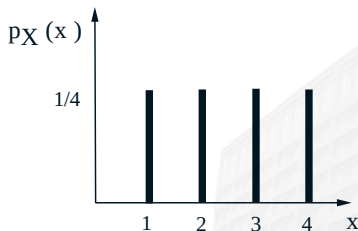
$$\mathbf{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Moreover, for a function $g(X)$, we have

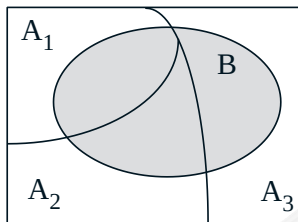
$$\mathbf{E}[g(X) \mid Y = y] = \sum_x g(x) p_{X|Y}(x|y)$$

Conditional PMF and expectation

- Let $A = \{X \geq 2\}$
 - $p_{X|A}(x) = 1/3$ for $x = 2, 3, 4$
 - $\mathbf{E}[X | A] = 3$



- Partition of sample space into disjoint events $A_1, A_2, \dots, A_n$



$$\mathbf{P}(B) = \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \cdots + \mathbf{P}(A_n)\mathbf{P}(B \mid A_n)$$

$$p_X(x) = \mathbf{P}(A_1)p_{X|A_1}(x) + \cdots + \mathbf{P}(A_n)p_{X|A_n}(x)$$

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X \mid A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X \mid A_n]$$

- X : number of independent coin tosses until first head

$$p_X(k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

- Expectation and variance of X :

$$\mathbf{E}[X] = \sum_{k=1}^{\infty} kp_X(k) = \sum_{k=1}^{\infty} k(1 - p)^{k-1}p$$

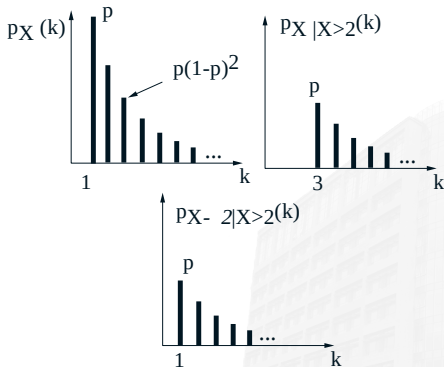
$$\mathbf{E}[X^2] = \sum_{k=1}^{\infty} k^2 p_X(k) = \sum_{k=1}^{\infty} k^2 (1 - p)^{k-1}p$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

- **Memoryless property:** Given that $X > 2$, the random variable $X - 2$ has same geometric PMF.

$$\begin{aligned} p_{X|X>2}(k) &= \frac{\mathbf{P}(\{X > 2\} \cap \{X = k\})}{\mathbf{P}(X > 2)} \\ &= \frac{(1-p)^{k-1}p}{1-p-p(1-p)} \\ &= (1-p)^{k-3}p, \quad k = 3, 4, \dots \end{aligned}$$

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- Define:

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- If the first toss fails, we are back where we started.

$$\begin{aligned}\mathbf{E}[X] &= \mathbf{P}(X = 1)\mathbf{E}[X \mid X = 1] + \mathbf{P}(X > 1)\mathbf{E}[X \mid X > 1] \\ &= p + (1 - p)(1 + \mathbf{E}[X])\end{aligned}$$

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- Solve to get $\mathbf{E}[X] = 1/p$

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$$\begin{aligned}\mathbf{E}[X^2] &= \mathbf{P}(X = 1)\mathbf{E}[X^2 \mid X = 1] + \mathbf{P}(X > 1)\mathbf{E}[X^2 \mid X > 1] \\ &= p + (1 - p)\mathbf{E}[(1 + X)^2] = p + (1 - p)(1 + 2\mathbf{E}[X] + \mathbf{E}[X^2])\end{aligned}$$

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- Then

$$\begin{aligned}\mathbf{E}[X^2] &= \frac{1 + 2(1 - p)\mathbf{E}[X]}{p} = \frac{2 - p}{p^2} \\ \text{var}(X) &= (1 - p)/p^2\end{aligned}$$

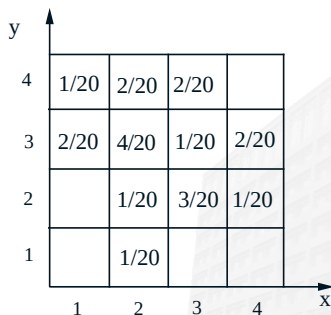
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Definition of Joint PMFs

The joint PMF $p_{X,Y}$ of X and Y is defined by

$$p_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y) = \mathbf{P}(\{X(\omega) = x\} \cap \{Y(\omega) = y\})$$



- $\sum_x \sum_y p_{X,Y}(x, y) = 1$

- **Marginal PMF:**

$$p_X(x) = \sum_y \mathbf{P}(X = x, Y = y) = \sum_y p_{X,Y}(x, y)$$

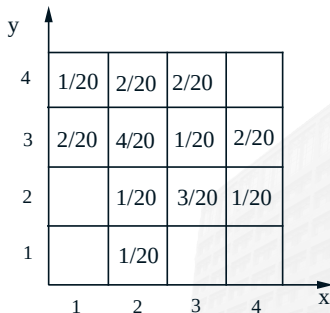
- **Conditional PMF:**

$$p_{X|Y}(x | y) = \mathbf{P}(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

- $\sum_x p_{X|Y}(x | y) = 1$

Marginal and Conditional PMFs

- $p_X(x) = ?$
- $p_Y(y) = ?$
- $p_{X|Y}(x|2) = ?$



- Function of multiple RVs: $Z = g(X, Y)$

$$p_Z(z) = \sum_{\{(x,y)|g(x,y)=z\}} p_{X,Y}(x, y)$$

- More than two RVs:

$$p_{X,Y,Z}(x, y, z) = \mathbf{P}(X = x, Y = y, Z = z)$$

and

$$p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z)$$

$$p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x, y, z)$$

- Recall that: $\mathbf{E}[X] = \sum_x xp_X(x)$. Furthermore,

$$\mathbf{E}[g(X, Y)] = \sum_x \sum_y g(x, y)p_{X,Y}(x, y)$$

$$\mathbf{E}[g(X, Y, Z)] = \sum_x \sum_y \sum_z g(x, y, z)p_{X,Y,Z}(x, y, z)$$

- In general: $\mathbf{E}[g(X, Y)] \neq g(\mathbf{E}[X], \mathbf{E}[Y])$
- $\mathbf{E}[\alpha X + \beta] = \alpha\mathbf{E}[X] + \beta$
- $\mathbf{E}[X + Y + Z] = \mathbf{E}[X] + \mathbf{E}[Y] + \mathbf{E}[Z]$

- X is the number of successes in n independent trials and p is probability of success. What is $\mathbf{E}[X]$?



- X is the number of successes in n independent trials and p is probability of success. What is $\mathbf{E}[X]$?

Solution:

- Define

$$X_i = \begin{cases} 1, & \text{if success in trial } i, \\ 0, & \text{otherwise.} \end{cases}$$

- Their sum

$$X = X_1 + X_2 + \cdots + X_n$$

- $\mathbf{E}[X] = \sum_{i=1}^n \mathbf{E}[X_i] = np$
- $\text{var}(X) = np(1 - p)$

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Definition of Independence

Two RVs X and Y are **independent** if

$$p_{X,Y}(x,y) = p_X(x) \cdot p_Y(y)$$

for all x and y .

- If X and Y are independent:

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)]$$

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

- Examples:

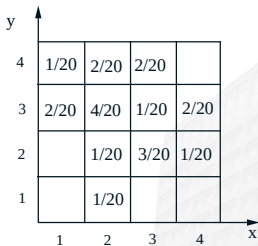
- If $X = Y$, $\text{var}(X + Y) = 4 \cdot \text{var}(X)$
- If X and Y are independent, $Z = X - 3Y$, $\text{var}(Z) = \text{var}(X) + 9 \cdot \text{var}(Y)$

Definition of Conditional Independence

Given a positive probability event A , two RVs X and Y are **conditional independent** if

$$p_{X,Y|A}(x,y) = p_{X|A}(x) \cdot p_{Y|A}(y)$$

for all x and y .



- Independent?
- What if we condition on $X \leq 2$ and $Y \geq 3$?

The hat problem

- n people throw their hats in a box and then pick one at random.
- X : number of people who get their own hat
- Find $\mathbf{E}[X]$



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- X : number of people who get their own hat
- Find $\mathbf{E}[X]$

Solution:

- Define:

$$X_i = \begin{cases} 1, & \text{if } i \text{ selects own hat} \\ 0, & \text{otherwise.} \end{cases}$$

- $X = X_1 + X_2 + \cdots + X_n$
- $\mathbf{E}[X_i] = \frac{1}{n}$
- Are the X_i independent?
- $\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_n] = 1$

- $\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \mathbf{E}[X^2] - 1$

$$X^2 = \sum_i X_i^2 + \sum_{i,j:i \neq j} X_i X_j$$

- $\mathbf{E}[X_i^2] = \frac{1}{n}$

$$\mathbf{P}(X_1 X_2 = 1) = \mathbf{P}(X_1 = 1) \cdot \mathbf{P}(X_2 = 1 \mid X_1 = 1)$$

- $\mathbf{E}[X^2] = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 2$

- $\text{var}(X) = 1$

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1. Consider an experiment in which a fair four-sided die (with faces labeled 0, 1, 2, 3) is thrown once to determine how many times a fair coin is to be flipped. In the sample space of this experiment, random variables N and K are defined by

- N = the result of the die roll
- K = the total number of heads resulting from the coin flips

- (a) Determine and sketch $p_N(n)$
- (b) Determine and tabulate $p_{N,K}(n, k)$
- (c) Determine and sketch $p_{K|N}(k | 2)$
- (d) Determine and sketch $p_{N|K}(n | 2)$

2. You write a software program over and over, and each time there is probability p that it works correctly, independent of previous attempts. What is the variance of X , the number of tries until the program works correctly?
3. A coin that has probability of heads equal to p is tossed successively and independently until a head comes twice in a row or a tail comes twice in a row. Find the expected value of the number of tosses.

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- Total expectation theorem

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X \mid A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X \mid A_n]$$

- Multiple discrete random variables

$$p_{X,Y}(x, y) = \mathbf{P}(X = x \text{ and } Y = y)$$

- Independence

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

- Readings: Section 2.5-2.7
- Assignment: Problem Set 5
- Next lecture: Continuous random variables